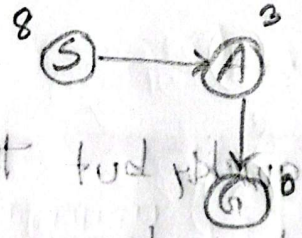
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Informed Search

Heuristic function — h(n)

estimate cost from a node n to goal node

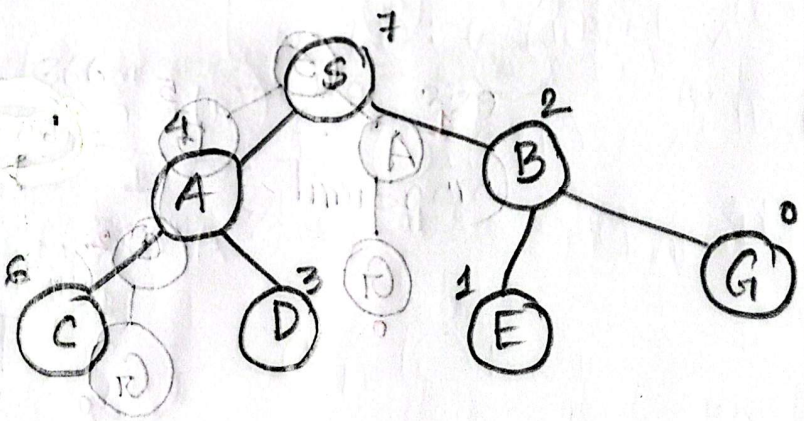


h(S) = 8 = S to G1 estimated cost is 8

h(A) = 3 = A to G1 estimated cost is 3

Greedy Search

explores the node with the lowest heuristic value first



S → B → G1

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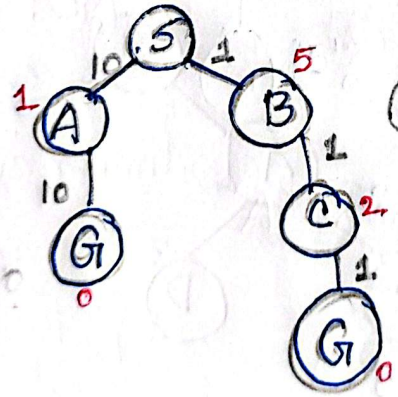
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Strategy :

expands the node that appears to be closest to the goal → it only uses heuristic value

Common case

reach the goal quickly but the path may not be shortest because it ignores actual path cost, heuristic value isn't always correct



greedy search

S → A → G

↳ cost = 10 + 10 = 20

but shortest path:

S → B → C → G

↳ cost = 1 + 1 + 1 = 3

"GOOD LUCK"

CamScanner

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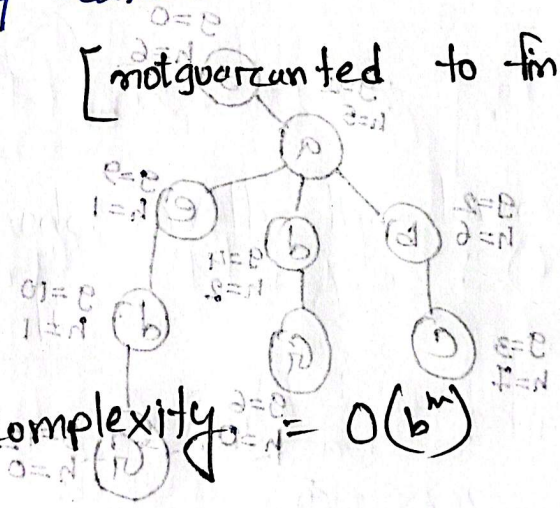
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Worst case :

Greedy search behave like poor-guided DFS if the heuristic value is missleading going Deep into the wrong path

Performance of Greedy Search

Complete ? No [not guarcanted to find solution]



Optimal ? no

Time and space complexity = $O(b^m)$

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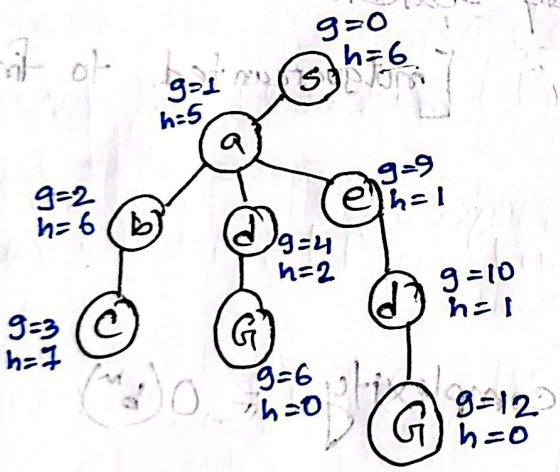
A* Search

↓

Combining UCS & Greedy

UCS → path cost → g(n)

Greedy → heuristic cost → h(n)



uniform search:

S → a → b → c → d → G → expansion of extra nodes

Greedy → S → a → e → d → G

A* search = UCS + greedy

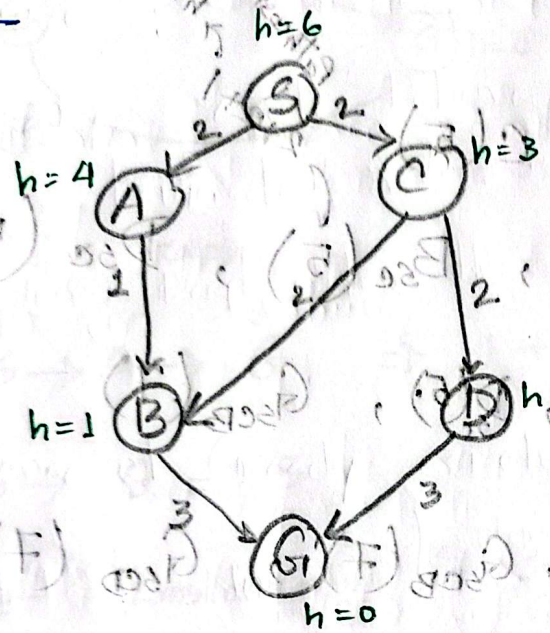
S → a → d → G

A* Search → f(n) = g(n) + h(n)

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

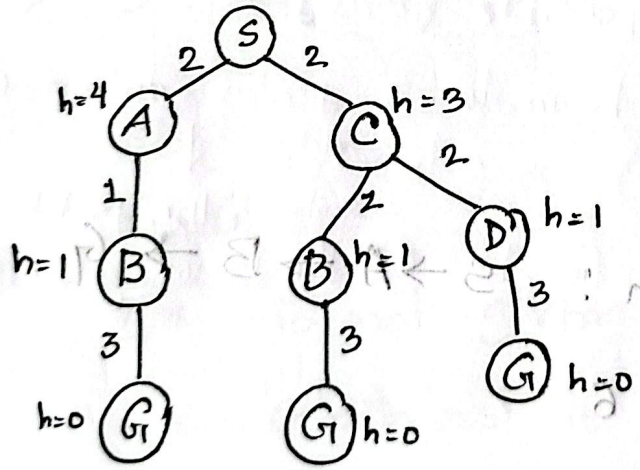
example



Show, expansion path
Solution path:
associated actual path cost for greedy best first search
A* tree search

Solve

converting in tree



Greedy

S: A(4), C(3)

$C_S(3)$: A(4), $B_{Se}(1)$, $D_{Se}(1)$

$B_{Se}(1)$: A(4), $D_{Se}(1)$, $G_{SeB}(0)$

solution path

$S \rightarrow C \rightarrow B \rightarrow G_1$

Cost: 2+2+3 = 7

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A* tree

$S: A(6), \quad C(5)$ (4+2)

$C_S(5): A(6), \quad B_{SC}(5), \quad D_{SC}(5)$ (Pathcost = 2+2+1 = value of B)

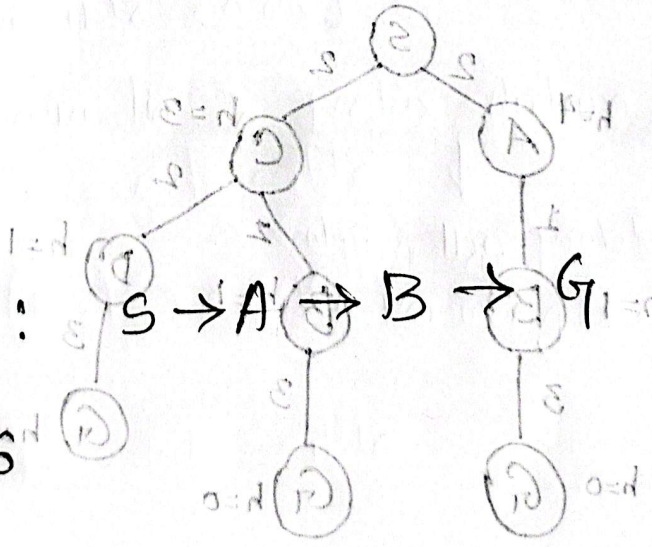
$B_{SC}(5): A(6), \quad D_{SC}(5), \quad G_{SCB}(7)$

$D_{SC}(5): A(6), \quad G_{SCB}(7), \quad G_{SCD}(7)$

$A_S(6): \quad G_{SCB}(7), \quad G_{SCD}(7), \quad B_{SA}(4)$ (2+1+1)

$B_{SA}(4): \quad G_{SCB}(7), \quad G_{SCD}(7), \quad G_{SAB}(6)$ (2+1+3+0)

$G_{SAB}(6)$

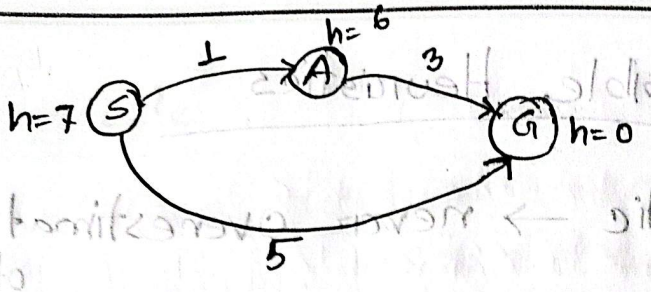


Solution Path : $S \rightarrow A \rightarrow B \rightarrow G_1$

Cost : 6

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A* always gives least cost → A* optimal

but for this graph,

$S \rightarrow A \rightarrow G \Rightarrow 1+3=4$

$S \rightarrow G = 5$

Path cost consideration $S \rightarrow A \rightarrow G$ optimal

but

$A^* \rightarrow S : A_S(7), G_S(5)$

G_S

$S \rightarrow G$

Here, A* is not giving optimal solution

$S \rightarrow G$ path cost is more than $S \rightarrow A \rightarrow G$

A* Optimal only when heuristic is Admissible

CamScanner

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Admissible Heuristics

Admissible heuristic → never overestimates the cost of goal

↓

guides in optimal way.

heuristic h is admissible if

$$0 \leq h(n) \leq h^*(n)$$

$h^*(n)$ → True cost to a nearest goal

A* search optimal? → Check admissible heuristic

admissible heuristic → Relax solution of a problem

↓

Solving the problem by removing restriction

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8 puzzle

Original rule → Only one adjacent tile can move into the empty space

Relaxed rule → A tile can move directly to its correct position

↳ Which is impossible practically

in 8 puzzle,

h_1 = count only the no of misplace tiles = 8

↳ more expansion → weaker heuristic

h_2 = count any tiles distance to goal = 18

↳ fewer expansion → stronger heuristic

Dominating heuristic → h_a when $h_a(n) \geq h_c$

↓

better heuristic → because → less node expansion

→ Faster search

→ Optimal

Guides A* well

b. Suppose you have two heuristic functions h1, h2 and h3. h1 and h2 are admissible but h3 is inadmissible. Consider the following heuristics and answer the questions: **[1+1.5]**

h4(n)=max(h1, h2, h3)

h5(n)=min(h1, h2, h3)

h6(n)=(h1+h2)/2

i. Which heuristic is inadmissible?

ii. Propose a heuristic that will be better than h4, h5 and h6.

iii. Suppose you have four heuristic functions h1, h2, h3, and h4. h1, h2, and h3 are admissible, but h4 is inadmissible. Consider the following heuristics and answer the questions:

- $h5(n)=h1^2+h2+h3$
- h6(n)=max(h1,h2,h3,h4)
- h7(n)=min(h1,h2,h4)
- h8(n)=(h1+h2+h3+h4)
- h9(n) = max(min(h1,h3),h4)

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TIME : DATE : / /

Practice Problem

h_1 = admissible

h_2 = admissible

h_3 = inadmissible

$h_4 = 0$ → admissible [actual cost is not -ve]

$h_5 = 2 \times h_2$ → possibly inadmissible

[might be greater than actual cost by multiplying by 2]

$h_6 = (h_1 + h_2)/2$ → admissible

[not more than actual cost as h_1 and h_2 are admissible so avg will be also admissible]

$h_7 = \max(h_1, h_2)$ → admissible

$\max(h_1, h_2) \leq$ actual cost

$h_8 = \min(h_1, h_3)$ → admissible

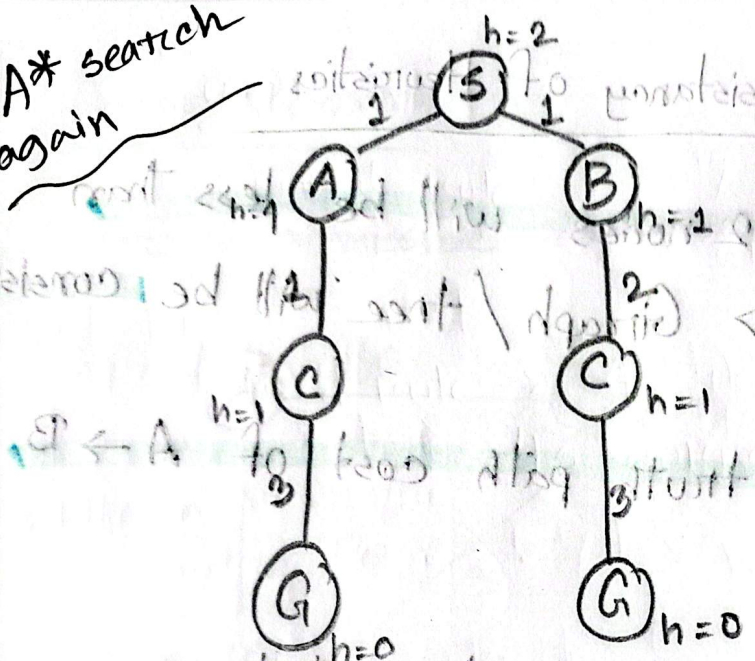
$h_9 = \max(h_2, h_3)$ → inadmissible

[h_3 is max value and not admissible]

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A* search again



S : $A_S(5)$, $B_S(2)$

$B_S(2)$: $A_S(5)$, $C_{SB}(4)$ (3+2+1)

$C_{SB}(4)$: $A_S(5)$, $G_{SBC}(6)$

$A_S(5)$: $G_{SBC}(6)$, $C_{SA}(3)$ → can't expand it anymore as it already visited

$G_{SBC}(6)$

$S \rightarrow B \rightarrow C \rightarrow G$

but not minimum cost

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Consistancy of Heuristics

difference of 2 nodes will be less than actual cost → Circaph / tree will be consistent

$h_A - h_B \leq$ truth path cost of $A \rightarrow B$

To be optimal, A* search have to be →

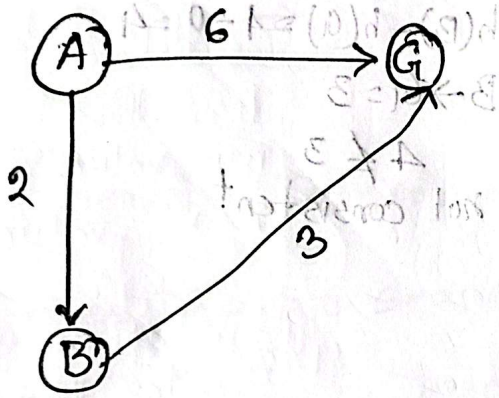
h → admissible and consistent both

TOPIC NAME : _____ DAY : _____

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Heuristic Testing

Detect admissible and consistent heuristics



h(A)	h(B)	h(G)	Admissible?	Consistent?
4	1	0	yes	NO
5	4	0	NO	NO
4	3	0	yes	yes
5	2	0	yes	NO

Admissible check

A → G = 6

A → B → G = 2+3 = 5 ← (choose min one)

given, h(A) = 4 ≤ 5 → A to G path cost 5

h(B) = 1 ≤ 3 → B to G path cost 3 and h(B) = 3

admissible

Consistency check

h(A) − h(B) = 4 − 1 = 3 ; A → B = 2

3 ≰ 2 — not

h(B) − h(G) = 1 − 0 = 1 ; B → G = 3

1 ≤ 3 → yes

here AND operation works one of them is not consistent so not consistent in general.

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· Admissible check

2nd case:

$A \to B \to G = 5$ (as this one shortest)

$h(A) = 5 \leq 5$ ✓

but $B \to G = 3$

so not admissible

consistency check

$h(A) - h(B) = 5 - 4 = 1$

$A \to B = 2$

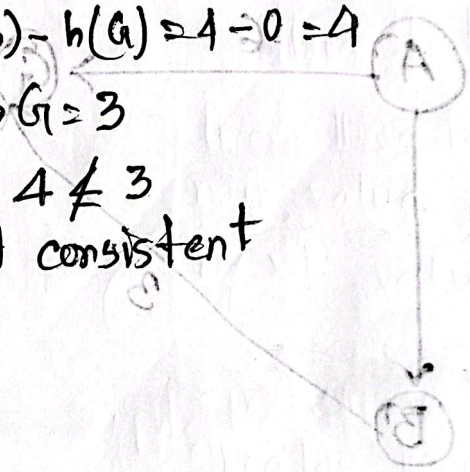
$1 \leq 2$ ✓

$h(B) - h(G) = 4 - 0 = 4$

$B \to G = 3$

$4 \not\leq 3$

not consistent



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Memory Bounded Informed Search

- IDA
- SMA
- RBFS

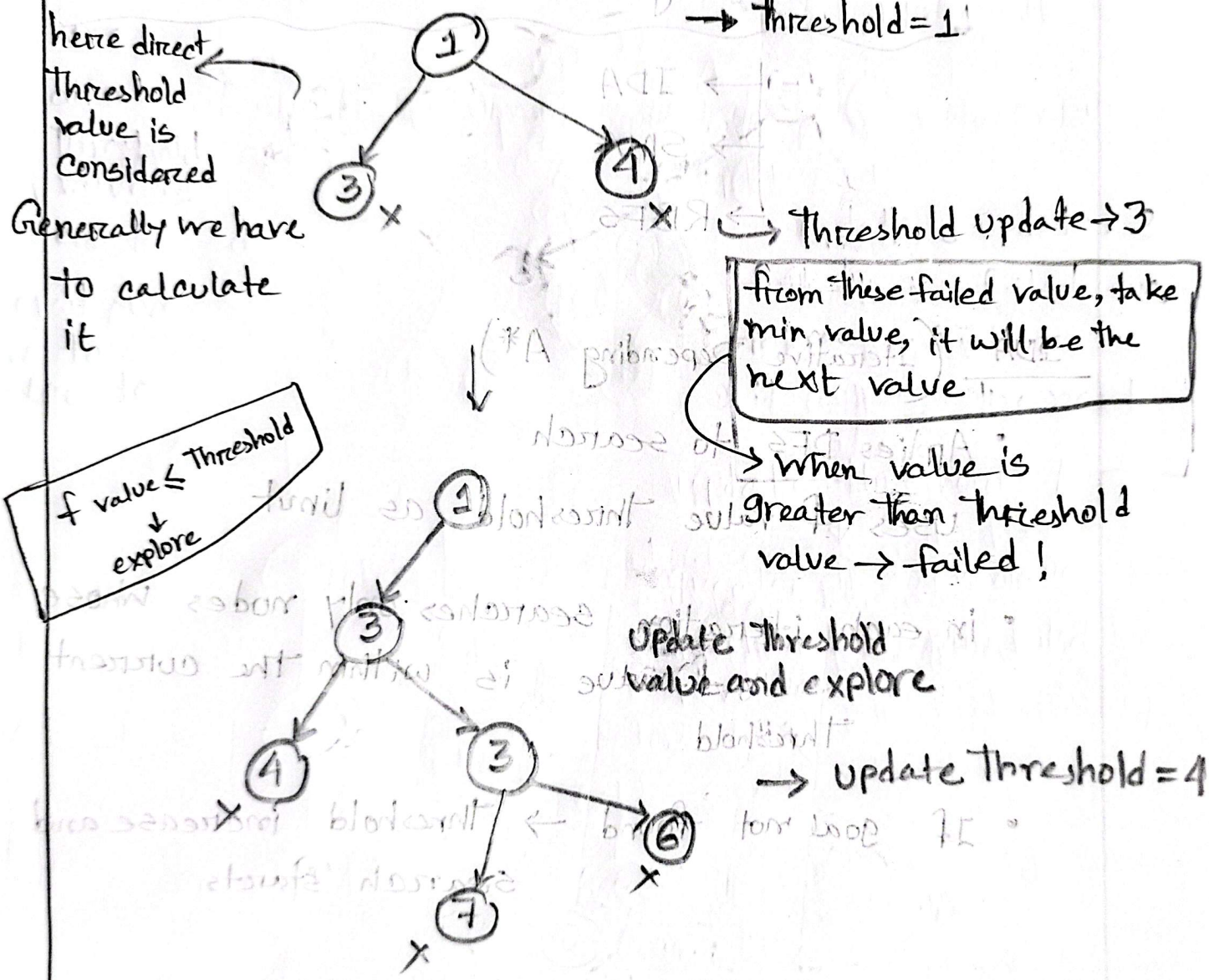
IDA (Iterative Depending A*)

- Applies DFS to search
- Uses threshold as limit
- in each iteration, searches only nodes whose f-value is within the current threshold
- If goal not found → threshold increase and search starts

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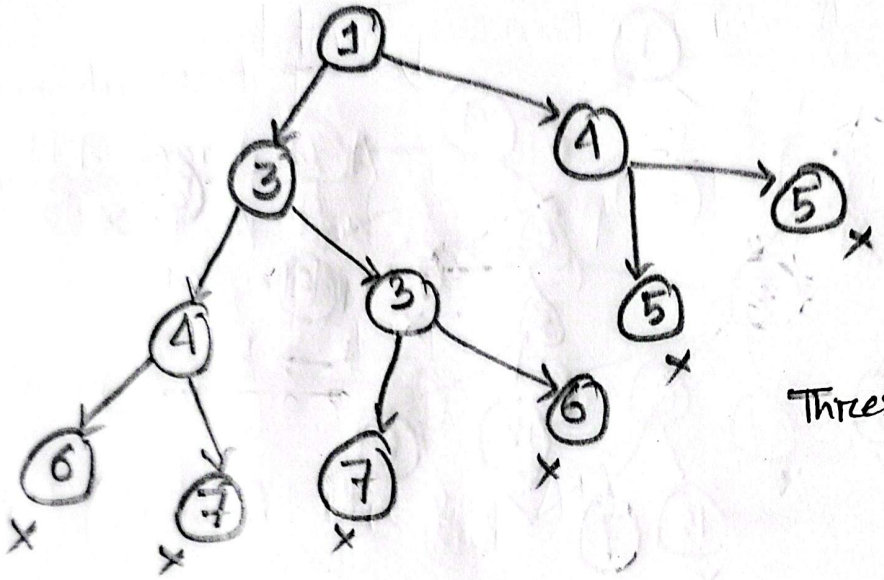
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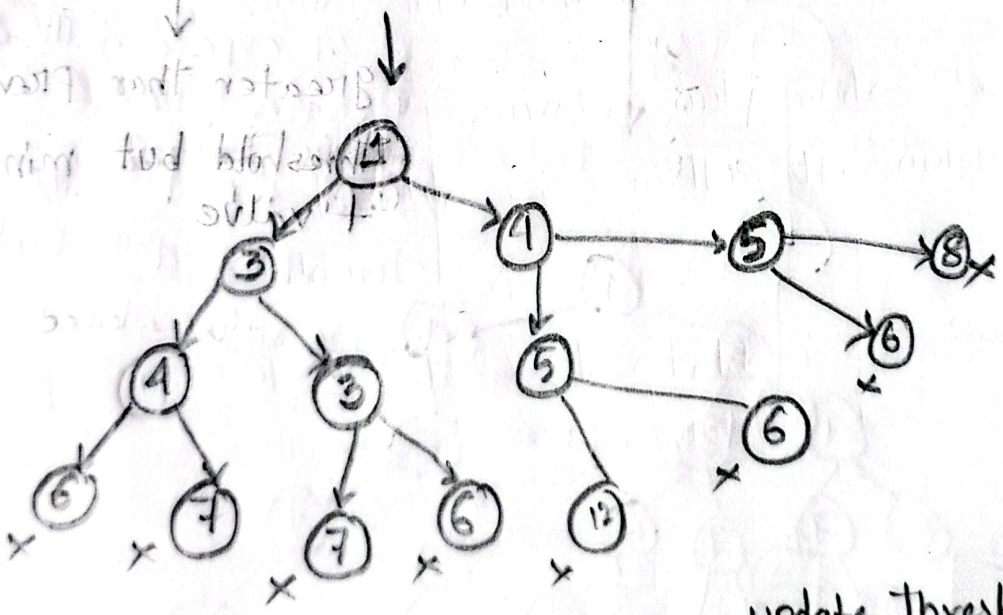
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Threshold update = 5

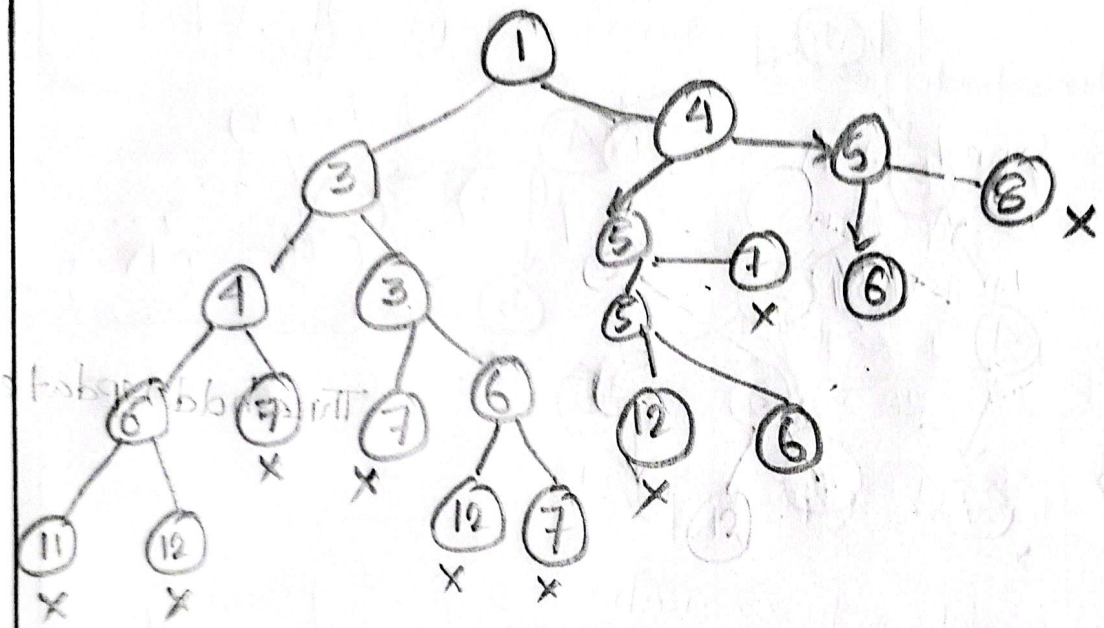


update threshold = 6

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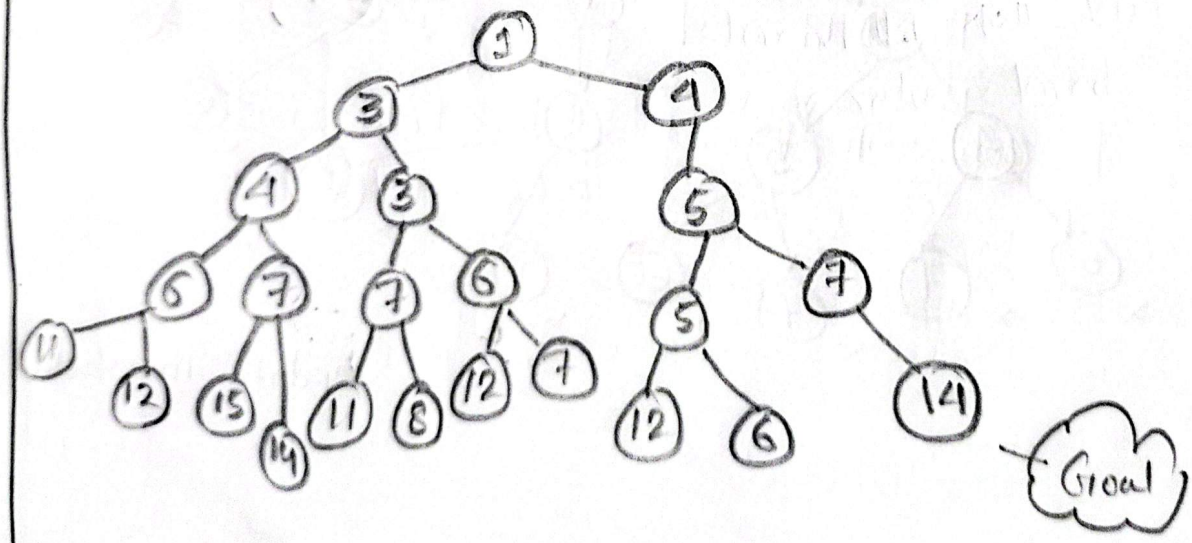
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update Threshold = 7

↓

greater than previous threshold but min failed f value



main concept :

Threshold থেকে কম/সমান f score এর node explore করব। যদি কোনো f score > threshold হয় তবে সেটা -failed. Then failed f-score এর মধ্যে min fscore কে threshold এ update করব and explore করব (এভাবে min f-score and explore করে goal খুজবো)

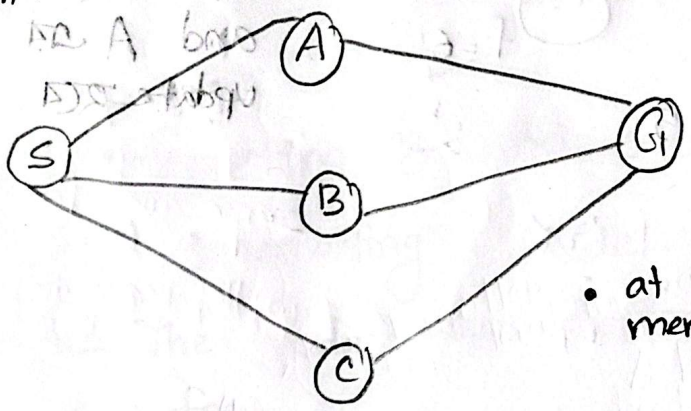
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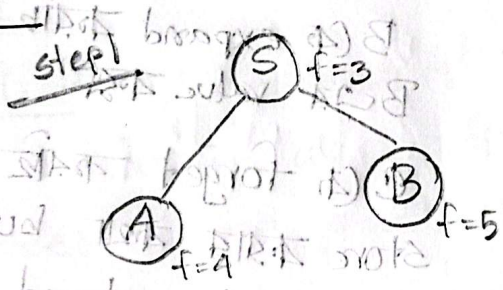
SMA* (Simplified Memory Bounded A*)

Main graph :

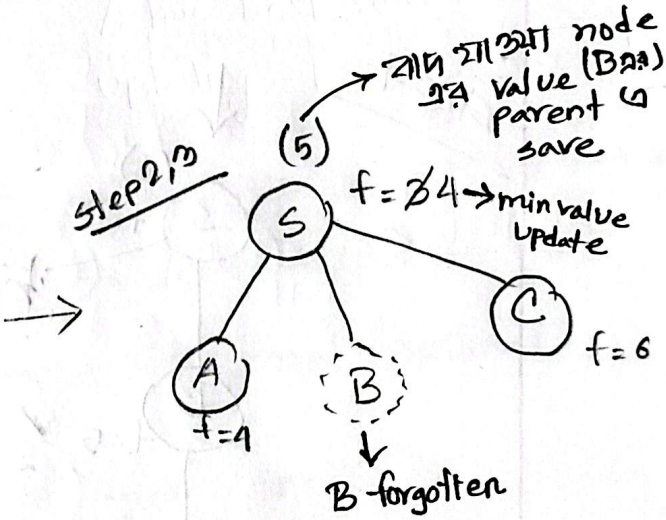


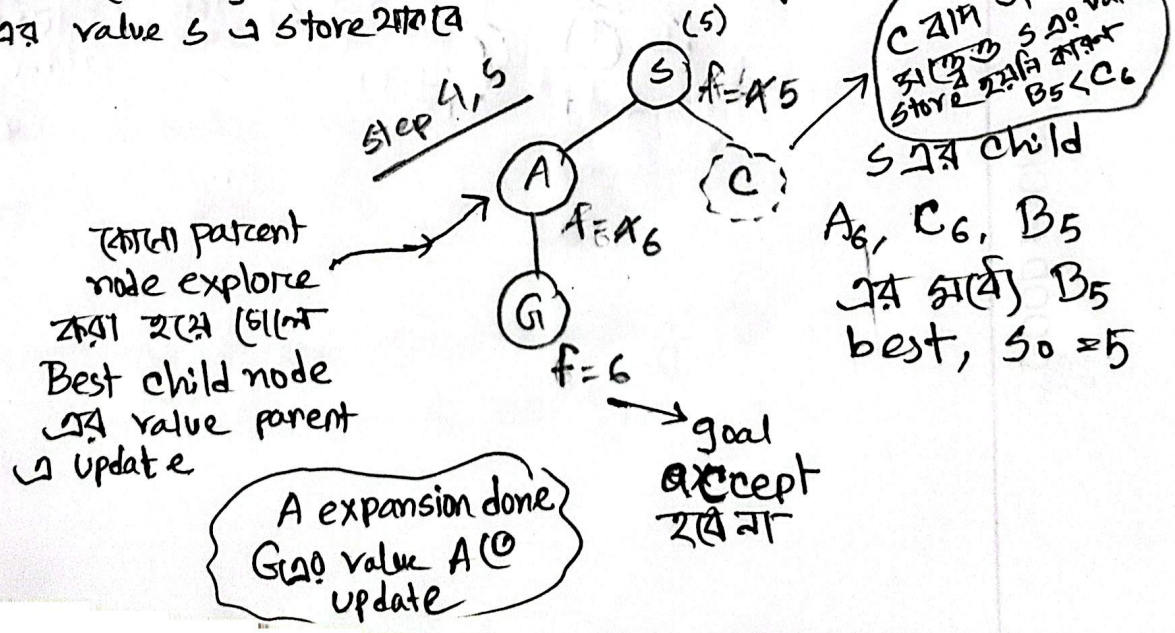
- at a time 3টা node রাখবে memory তে

SMA*



এখানে S এর f value A এর value = 4 update হবে কারণ এটা promissing value. আর B কে forget করব তবে B এর value S এ store রাখবে





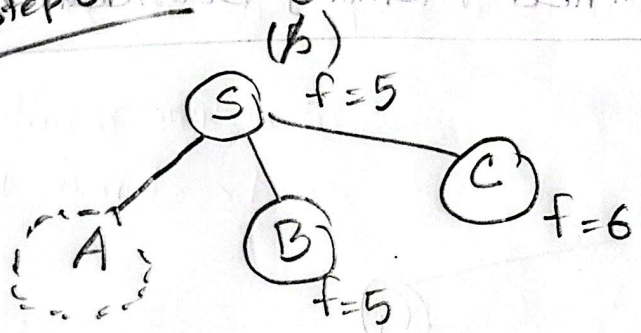
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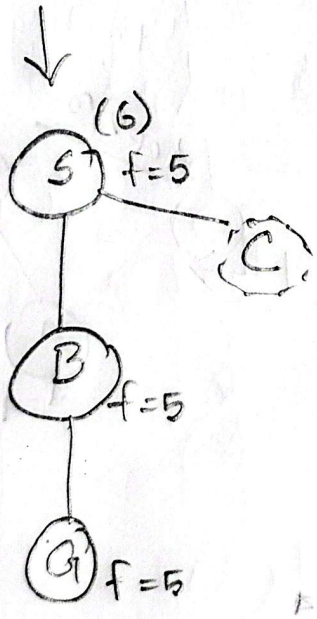
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Step 6.



A কে forget করব and A এর value S-এ store হবে

Step 7



B কে expand করছি করে B এর value করব

C কে forget করছি এর value store করার কথা but already same value stored

TOPIC NAME : all steps explained DAY :

TIME : DATE : / /

Step 1 : A = 4, B = 5 [considering f-value]

↓

A best and more promising

Step 2, 3 : • forgetting B and storing B's f-score at S. generating C node

• as all the nodes are explored by S, so f value of S is 4 → from A

Step 4, 5 : explore A and got G, update A's f-value to 6 and G is also 6. but this G is not accepted as goal because it's not optimal.

As memory full, so -forgetting C

but the value of C won't be stored

$B_5 < C_6$

again, f value of S will be updated to 5 from B as A_6, C_6 and B_5

Step 6 : forgetting A and regenerating B.

storing the value of A at S.

TOPIC NAME : _____ DAY : _____

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Step 7

explore B and found G ↓

and forgetting C as memory full

but C value is not storing its same goes for f-value of already stored (same value)

S will remain same

$S \rightarrow B \rightarrow G$ cost = 5

TOPIC NAME : _____ DAY : _____

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RBFS (recursive Best first Search)

$S.f = \max(S.g + S.h, \text{Parent}.f)$

- $S.f$: f-value of any node
- $S.g + S.h$: Self f-value
- $\text{Parent}.f$: f value of parent

Self f value and parent f value compare

* Best successor = Whose f-value is minimum

* if the best f value is greater than limit → don't expand

f-value of best node > f.limit → don't expand (failed)

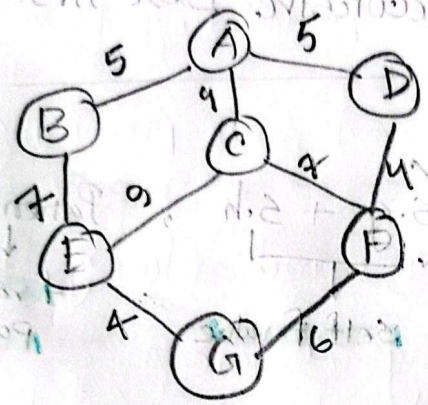
Otherwise explore

$f.\text{Limit} = \min(f.\text{limit}, \text{alt}.f)$

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

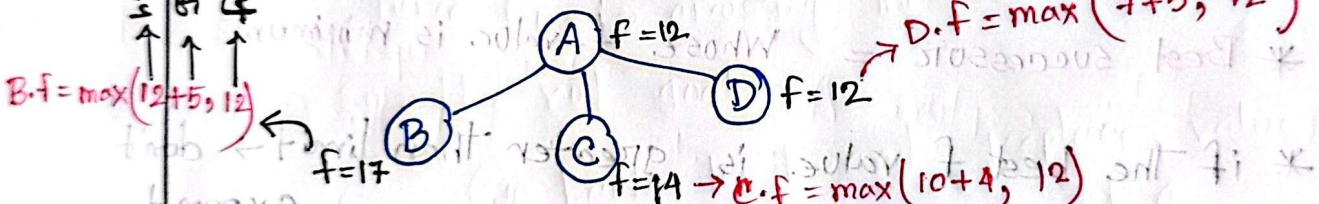
RBFS :

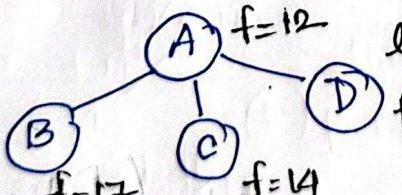


h value

A = 12
B = 12
C = 10
D = 7
E = 2
F = 6
G = 0

limit = ∞





limit = min(∞, 14) = 14

here best node is D=12

alterative best node is C = 14

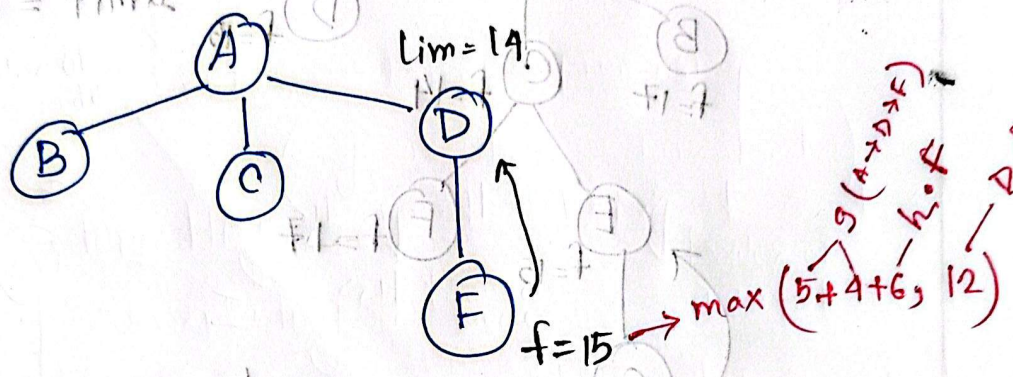
so, limit = min(∞, 14) (f.limit, alt.f)

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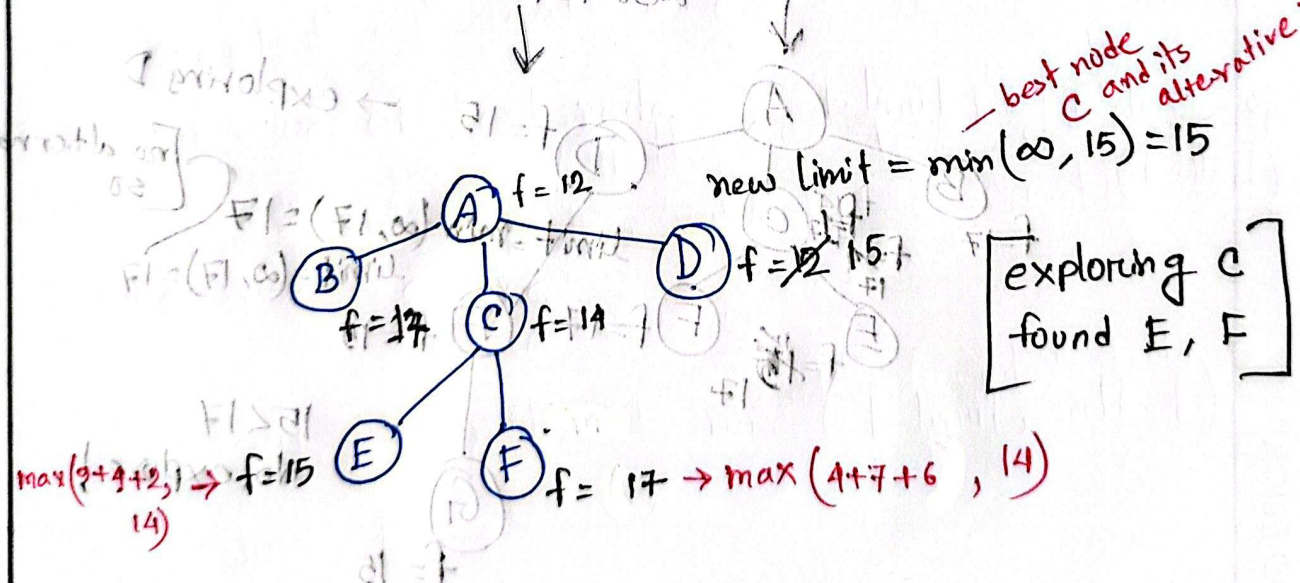
TOPIC NAME : _____ DAY : _____

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As, D.f doesn't cross the limit so let's explore D



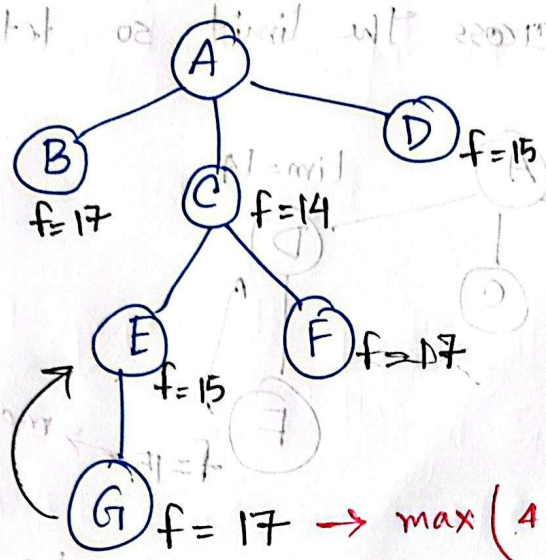
f value of F is crossing the limit (15 > 14) don't explore B to back to D and replace the value



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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /



limit = 15

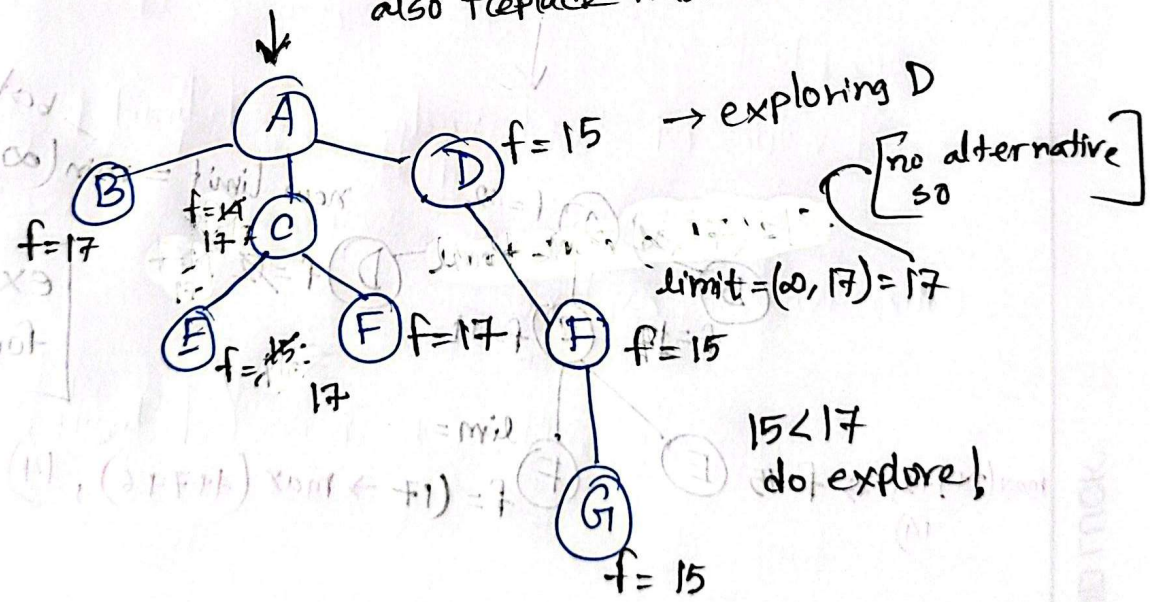
G f = 17 → max (4+9+4+0, 15) = 17

here G is crossing the limit 17 > 15

So, it's not goal.

Go back to E and replace value

also replace value of C



→ exploring D

[no alternative so limit = (∞, 17) = 17]

15 < 17 do explore

A → D → F → G cost = 15

GOOD LUCK™